

ENTERPRISE NETWORK MONITORING USING SPLUNK SIEM

*A Comprehensive Implementation Report on Deploying Splunk Enterprise
as a Security Information and Event Management (SIEM) Solution
for Real-Time Log Collection, Threat Detection, and Security Dashboard Visualization*

Prepared By:

Adeolu Olatundun Ogunbiyi

Cybersecurity Analyst

Github: github.com/adeoluogunbiyi || **LinkedIn:** [https:// www.linkedin.com/in/adeolu-ogunbiyi-928a9b4b/](https://www.linkedin.com/in/adeolu-ogunbiyi-928a9b4b/)

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1. Executive Summary

This report documents the end-to-end implementation of Splunk Enterprise as a Security Information and Event Management (SIEM) solution within a virtualized enterprise network environment. The project was designed to simulate a real-world Security Operations Center (SOC) deployment, demonstrating how centralized log collection, real-time event correlation, and security dashboard visualization can be achieved using industry-standard tools and methodologies.

The implementation encompassed the deployment of Splunk Enterprise 9.4.1 on a Windows Server 2022 virtual machine, the installation and configuration of the Splunk Universal Forwarder on a monitored endpoint (Windows 10/11), Windows Defender Firewall rule configuration to permit log forwarding traffic, and the creation of custom SPL (Search Processing Language) queries and interactive security dashboards for authentication monitoring and threat analysis.

The deployment successfully ingested over 5,400 Windows Security and System events from the monitored endpoint, enabling the creation of a custom Authentication Dashboard that provides real-time visibility into failed logon attempts (EventCode 4625), successful authentication events (EventCode 4624), and system-level setup activities (EventCode 1). This project demonstrates practical competency in SIEM deployment, log pipeline architecture, and security event visualization techniques that are directly applicable to enterprise SOC environments.

2. Project Objectives

The primary objectives of this project were established to align with industry best practices for SIEM deployment in enterprise environments. Each objective was designed to address a specific component of the security monitoring pipeline, from data collection through visualization and analysis.

- **Deploy Splunk Enterprise:** Install and configure Splunk Enterprise 9.4.1 as the central SIEM platform on a Windows Server 2022 environment, establishing the receiving indexer for log ingestion.
- **Configure Universal Forwarder:** Deploy and configure the Splunk Universal Forwarder (UF) on a monitored Windows endpoint to enable secure, real-time log forwarding to the Splunk indexer.
- **Establish Network Communication:** Configure Windows Defender Firewall inbound and outbound rules on both the server and endpoint to permit Splunk communication over the designated receiving port (TCP 9997).
- **Validate Data Ingestion:** Verify end-to-end log flow from the Windows endpoint through the Universal Forwarder to the Splunk indexer, confirming successful data indexing.
- **Develop SPL Search Queries:** Create targeted SPL queries to search, filter, and analyze ingested security events, including authentication logs, system events, and event code correlation.
- **Build Security Dashboards:** Design and implement interactive Splunk dashboards that provide SOC-ready visualization of security events, including failed logon trends, event timelines, and host-based activity summaries.

3. Scope and Environment

3.1 Lab Architecture

The project was implemented within an Oracle VirtualBox virtualization environment, simulating a segmented enterprise network. This architecture mirrors real-world deployment patterns where a centralized SIEM server receives forwarded logs from distributed endpoints across the network.

Component	Specification	Role
Hypervisor	Oracle VirtualBox 7.x	Virtualization Platform
SIEM Server	Windows Server 2022 Standard	Splunk Enterprise Indexer
Monitored Endpoint	Windows 10/11 Desktop	Log Source (Universal Forwarder)
Splunk Version	Splunk Enterprise 9.4.1	SIEM Platform
Forwarder Version	Splunk Universal Forwarder 9.4.3	Data Collection Agent
Network	Internal NAT / Host-Only	VM Communication
Receiving Port	TCP 9997	Splunk Data Forwarding
Web Interface	TCP 8000	Splunk Management Console
Host Server	WIN-CJ0J0L41QG0	Windows Server Hostname
Endpoint Host	DESKTOP-EPLK670	Monitored Endpoint Hostname

4. Technology Stack

Technology	Version / Type	Purpose
Splunk Enterprise	9.4.1	Central SIEM platform for log indexing, search, and visualization
Splunk Universal Forwarder	9.4.3	Lightweight agent for secure log collection and forwarding
Windows Server 2022	Standard Evaluation	SIEM server operating system
Windows Defender Firewall	Advanced Security	Network traffic filtering and rule management
Oracle VirtualBox	7.x	Hypervisor for lab virtualization
SPL (Search Processing Language)	Native	Query language for Splunk data analysis
Windows Event Logs	Security, System, Setup	Primary data sources for SIEM ingestion

5. Implementation Methodology

The implementation followed a structured, phased approach aligned with the NIST Cybersecurity Framework and Splunk deployment best practices. Each phase was executed sequentially, with validation checkpoints to confirm successful completion before proceeding to the next stage.

5.1 Phase 1: Splunk Enterprise Installation

The first phase involved downloading and installing Splunk Enterprise 9.4.1 on the Windows Server 2022 virtual machine. Splunk Enterprise serves as the central indexer and search head in this deployment, responsible for receiving, parsing, indexing, and storing all forwarded log data.

5.1.1 Download and Preparation

The Splunk Enterprise installer was downloaded directly from the official Splunk website (splunk.com). The download was verified for integrity before proceeding with installation. The screenshot below confirms the successful completion of the Splunk Enterprise download.

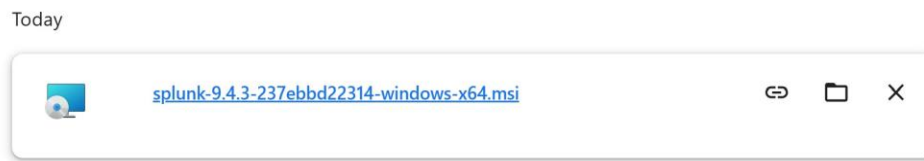


Figure 1. *Splunk Enterprise installer successfully downloaded to the Windows Server 2022 environment, ready for deployment.*

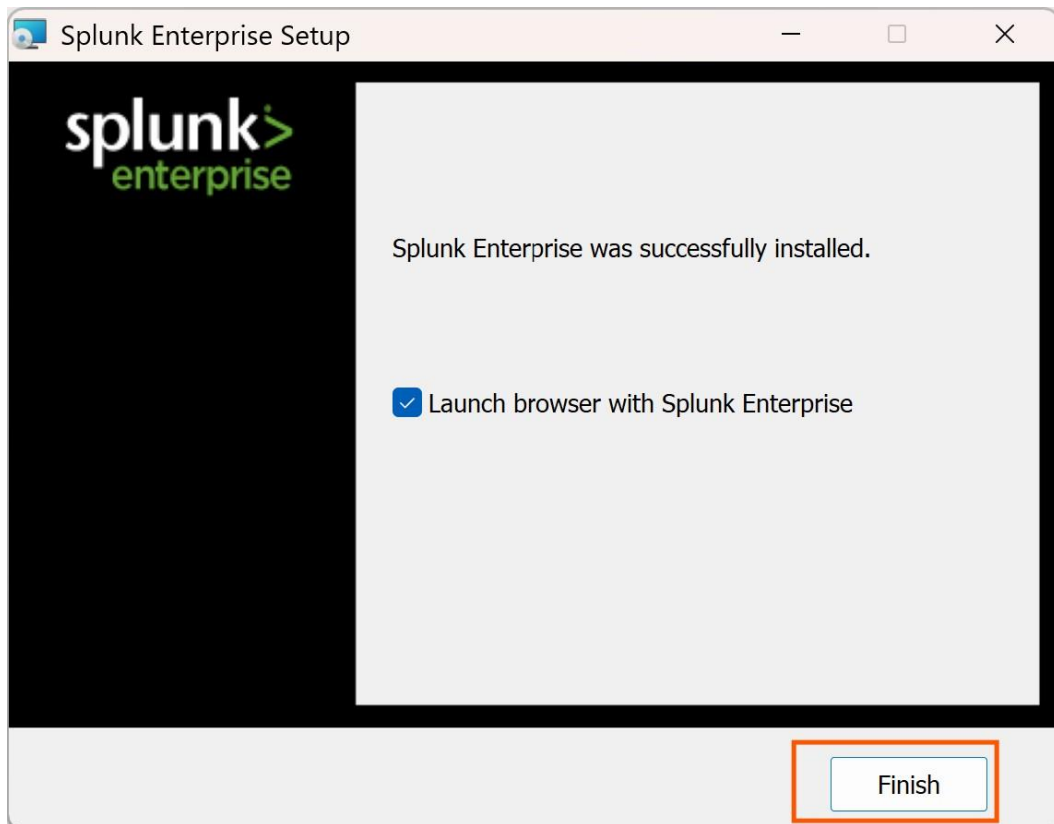
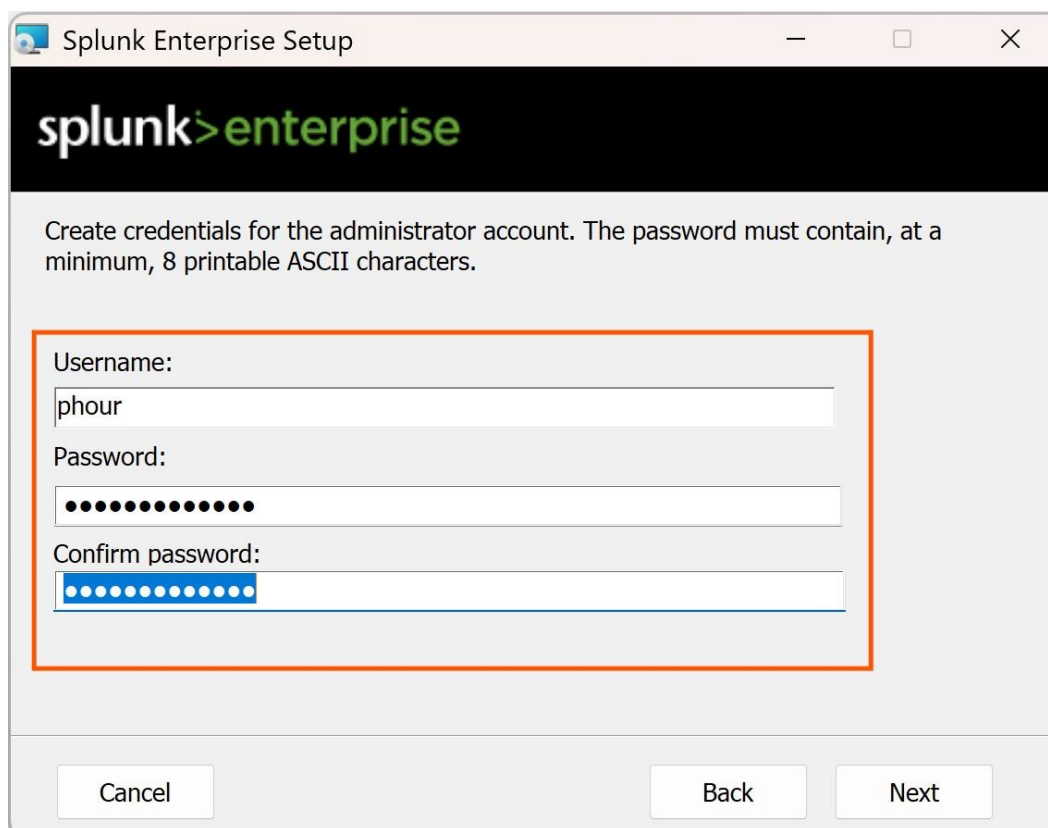


Figure 2. Download completion confirmation showing the Splunk Enterprise installation package fully retrieved and verified.

5.1.2 Installation and User Configuration

The installation process was initiated by executing the Splunk Enterprise MSI installer with administrative privileges. During installation, a dedicated Splunk administrator account was created with a secure password. The installer configured the Splunk service to run as a local system account, ensuring persistent operation across system restarts.



The screenshot shows a Windows-style window titled "Splunk Enterprise Setup". The window has a black header bar with the "splunk>enterprise" logo in white and green. Below the header, a message reads: "Create credentials for the administrator account. The password must contain, at a minimum, 8 printable ASCII characters." The main content area contains three input fields, all enclosed in an orange rectangular border:

- Username:** A text input field containing the text "phour".
- Password:** A password input field filled with 12 black dots.
- Confirm password:** A password input field filled with 12 blue dots.

At the bottom of the window, there are three buttons: "Cancel", "Back", and "Next".

Figure 3. *Splunk Enterprise installation wizard displaying the administrator user creation interface. A dedicated 'phour44' account was configured with appropriate credentials for SIEM management.*

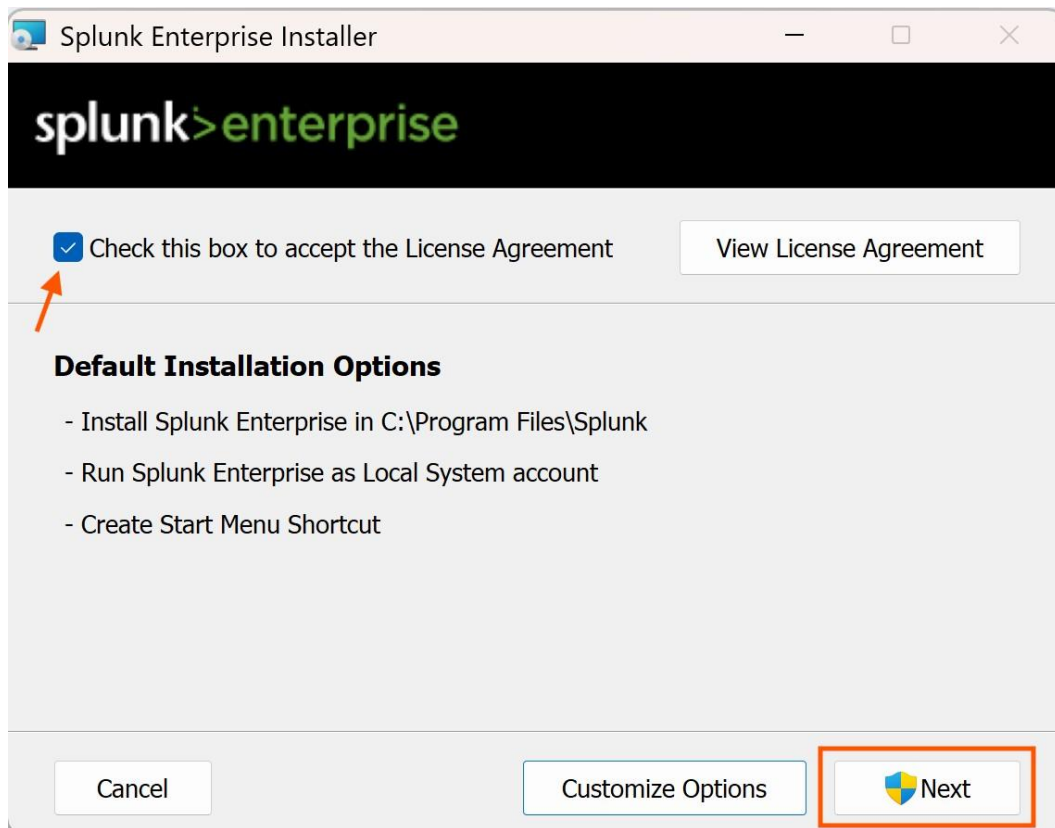


Figure 4. *Splunk Enterprise installation in progress on Windows Server 2022. The installer is deploying core Splunk services, indexer components, and the web-based management interface.*

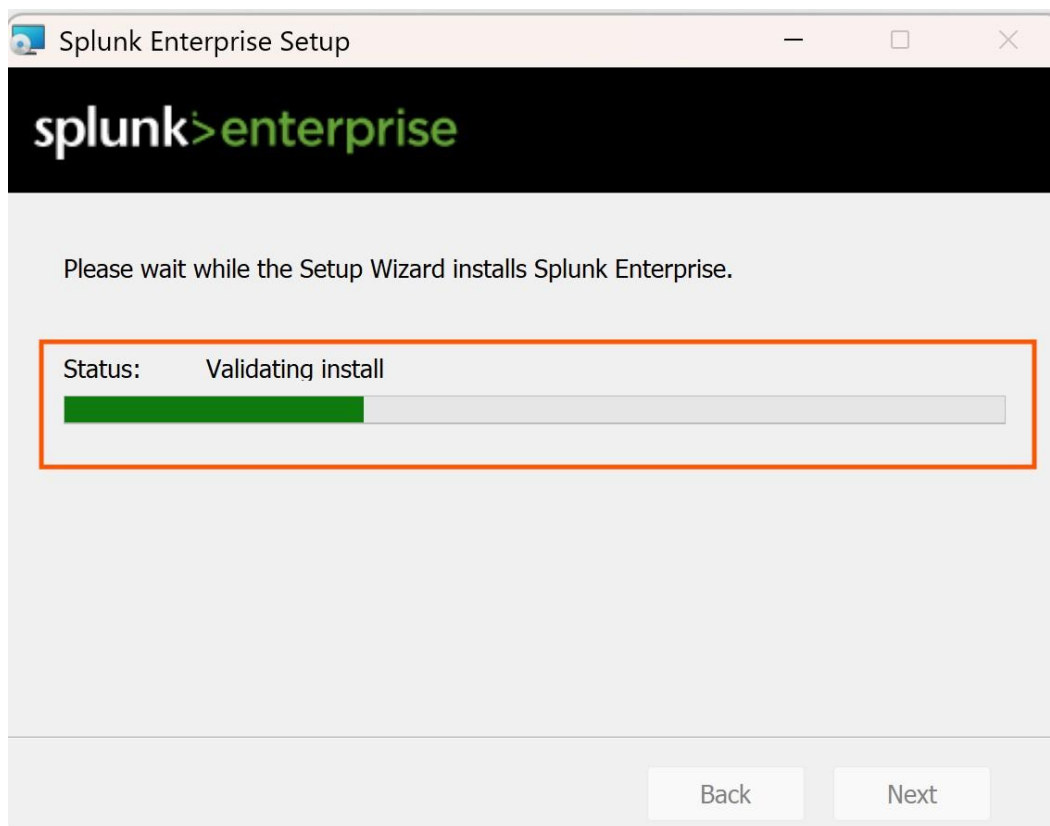
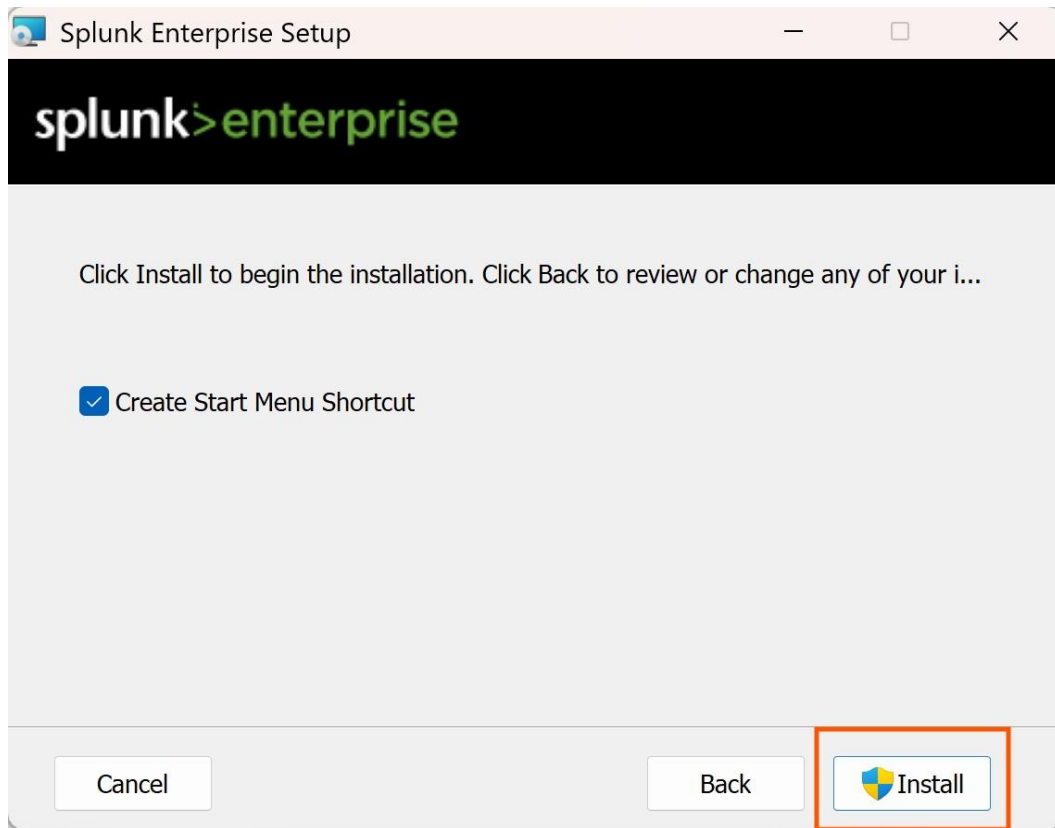


Figure 5. *Splunk Enterprise installation progress indicator showing component extraction and service registration on the server.*



***Figure 6.** Pre-installation summary screen confirming deployment parameters before final Splunk Enterprise installation execution.*

5.1.3 Licensing and Activation

Upon successful installation, Splunk Enterprise was launched and the licensing agreement was accepted. The platform was activated under a Splunk Enterprise trial license, which provides full functionality for evaluation purposes. The licensing screen confirmed successful activation and readiness for configuration.

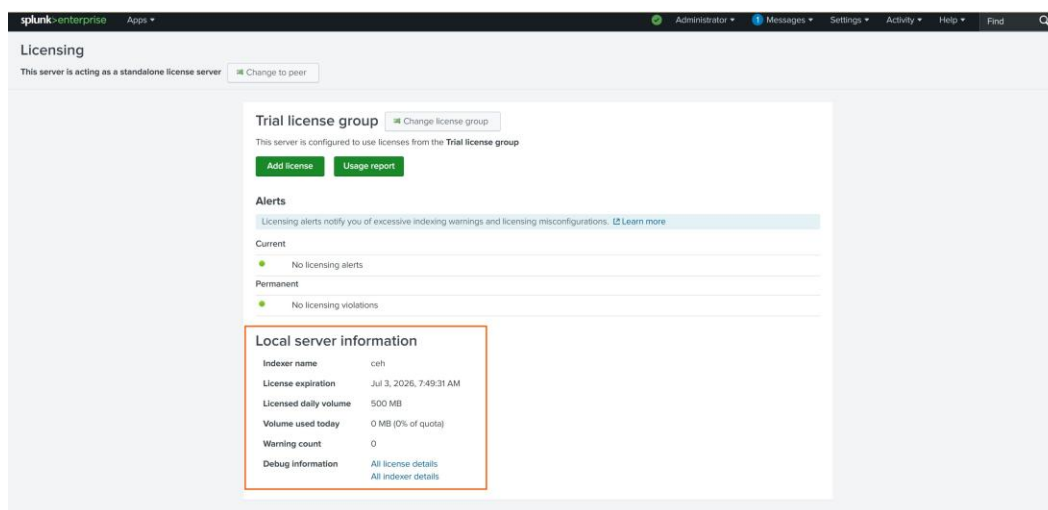


Figure 7. Splunk Enterprise licensing agreement screen presented during initial platform activation and configuration.

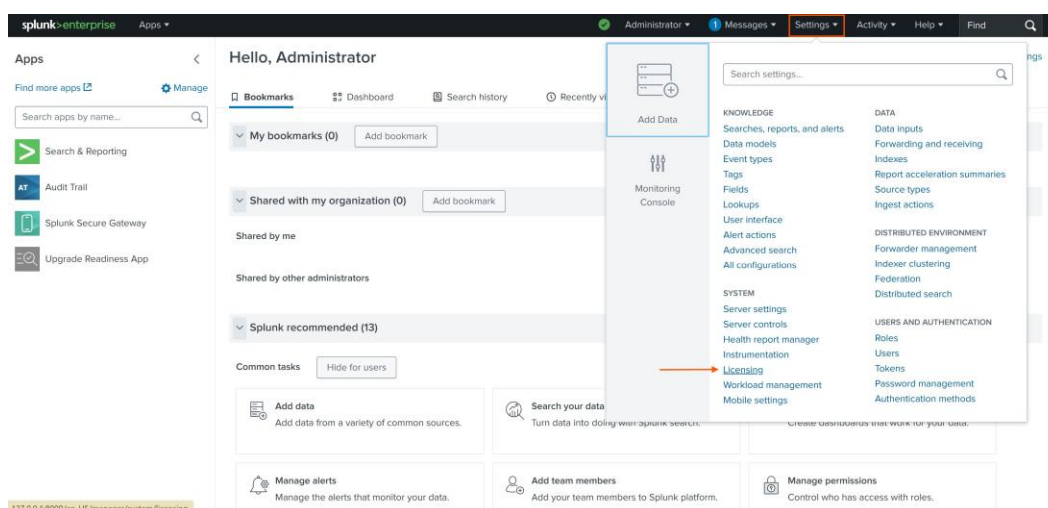


Figure 8. Splunk licensing confirmation page displaying the accepted license terms and activation status for the enterprise deployment.

5.1.4 Web Interface Access

After successful installation and licensing, the Splunk Enterprise web interface was accessed at <http://127.0.0.1:8000>. The login page was presented with the previously configured administrator credentials (username: phour44). Successful authentication confirmed that the Splunk web interface was operational and ready for configuration.

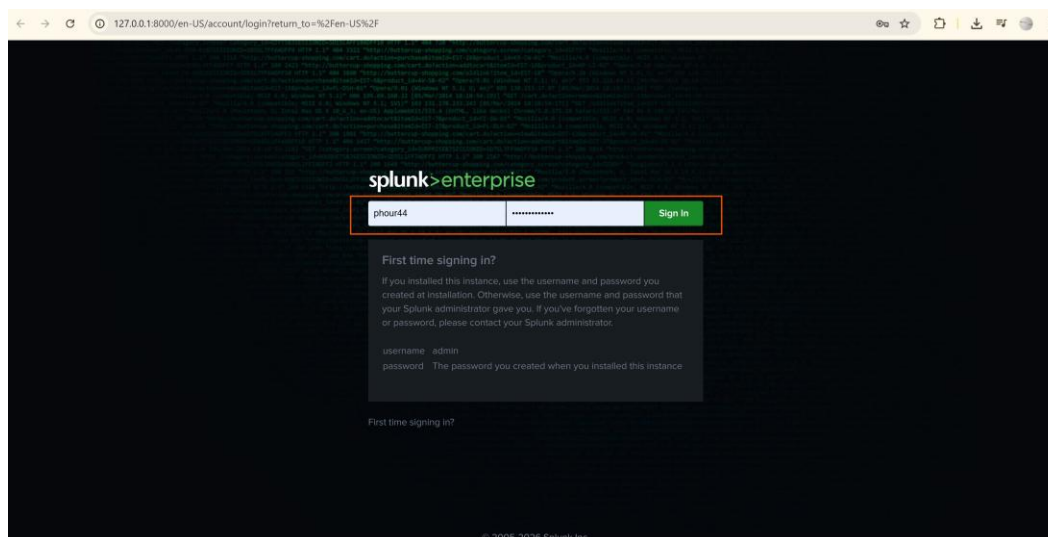


Figure 9. Splunk Enterprise web login interface accessed at 127.0.0.1:8000. The administrator account 'phour44' is being used to authenticate into the SIEM management console.

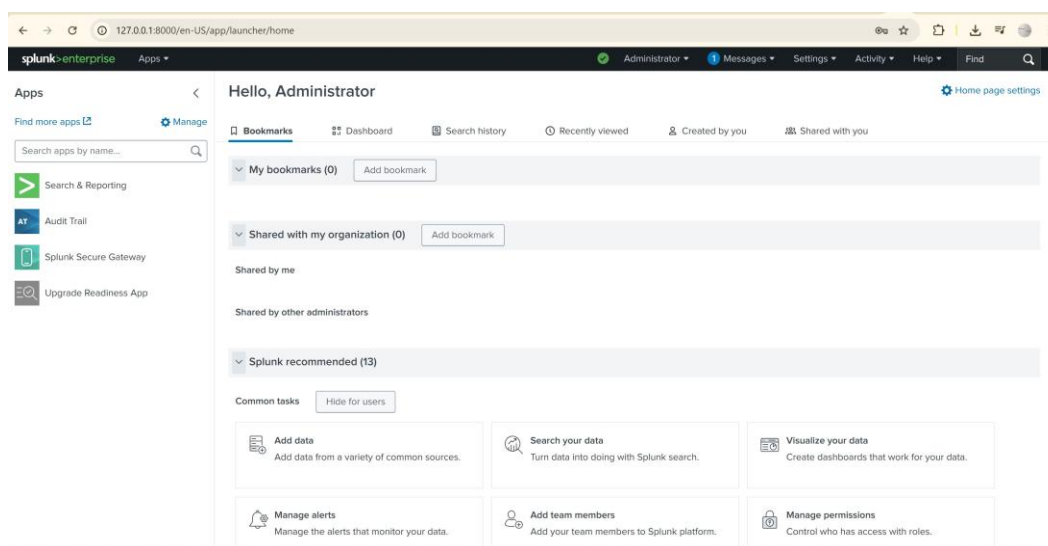


Figure 10. Splunk Enterprise home dashboard after successful authentication, displaying the main navigation interface with access to Search & Reporting, Audit Trail, and system management applications.

5.2 Phase 2: Splunk Universal Forwarder Deployment

The Splunk Universal Forwarder (UF) is a lightweight, dedicated agent designed to collect and forward log data from monitored endpoints to the central Splunk indexer. Unlike heavy forwarders, the UF has a minimal resource footprint and does not include a search or web interface, making it ideal for deployment across large numbers of endpoints in enterprise environments.

5.2.1 Forwarder Installation

The Universal Forwarder was installed on the monitored Windows endpoint (DESKTOP-EPLK670) using the standard MSI installer. The installation wizard guided the configuration of the receiving indexer's IP address and the designated receiving port (TCP 9997), establishing the communication channel between the endpoint and the Splunk server.

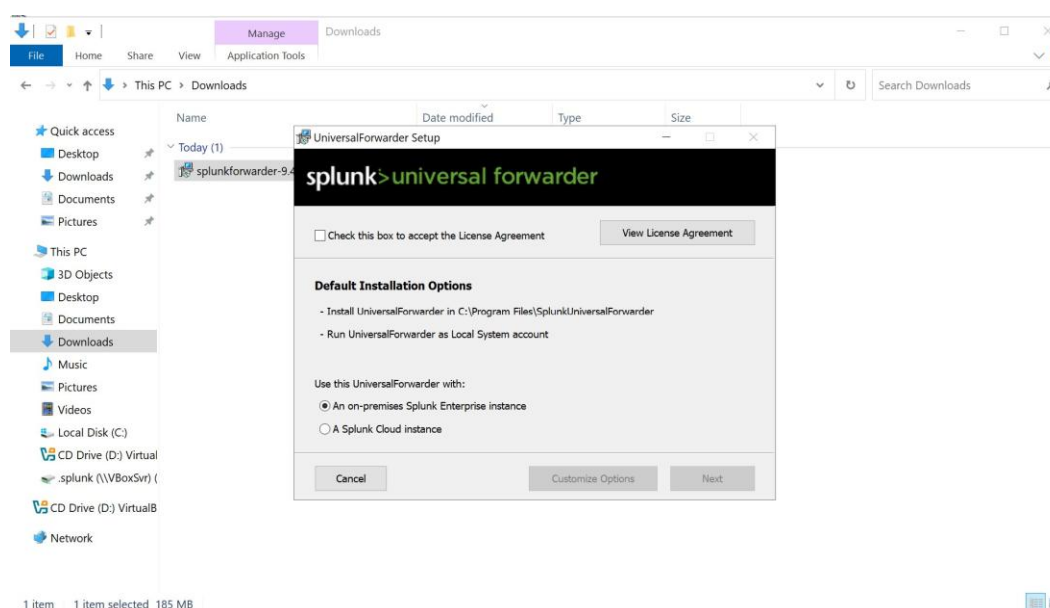


Figure 11. Splunk Universal Forwarder installation wizard launched on the monitored endpoint (DESKTOP-EPLK670), initiating the deployment of the log collection agent.

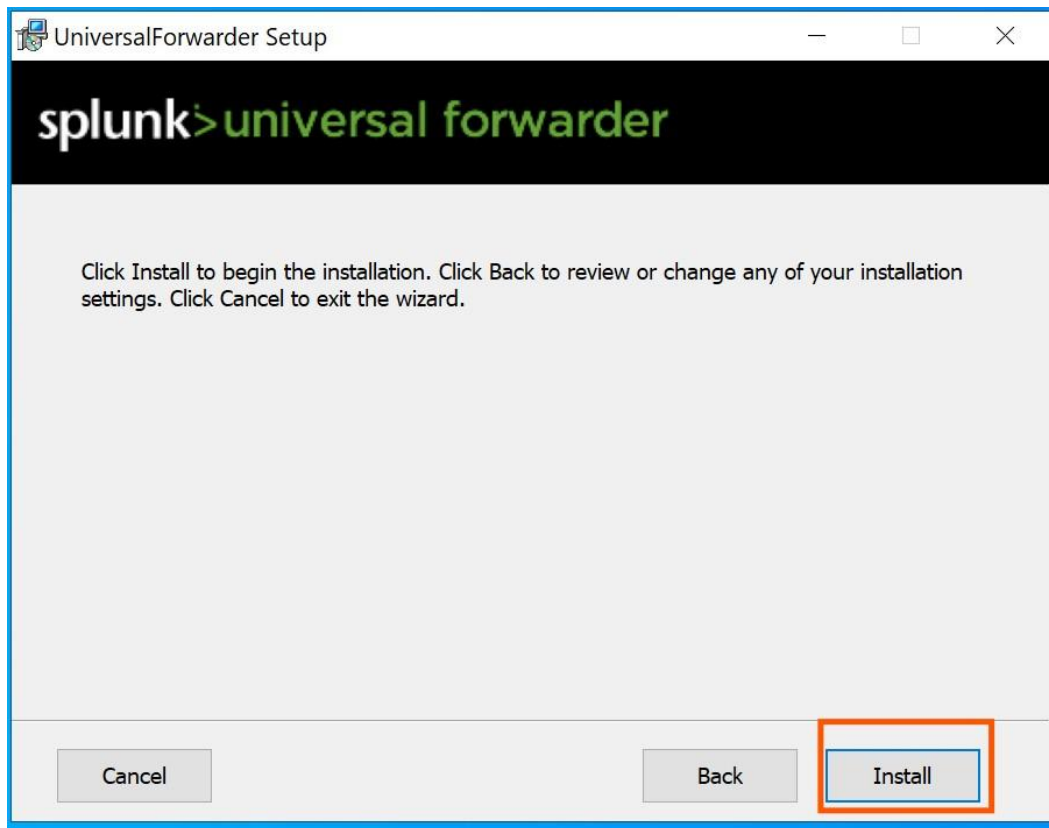


Figure 12. *Universal Forwarder pre-installation summary confirming component selection and deployment path before final installation execution.*

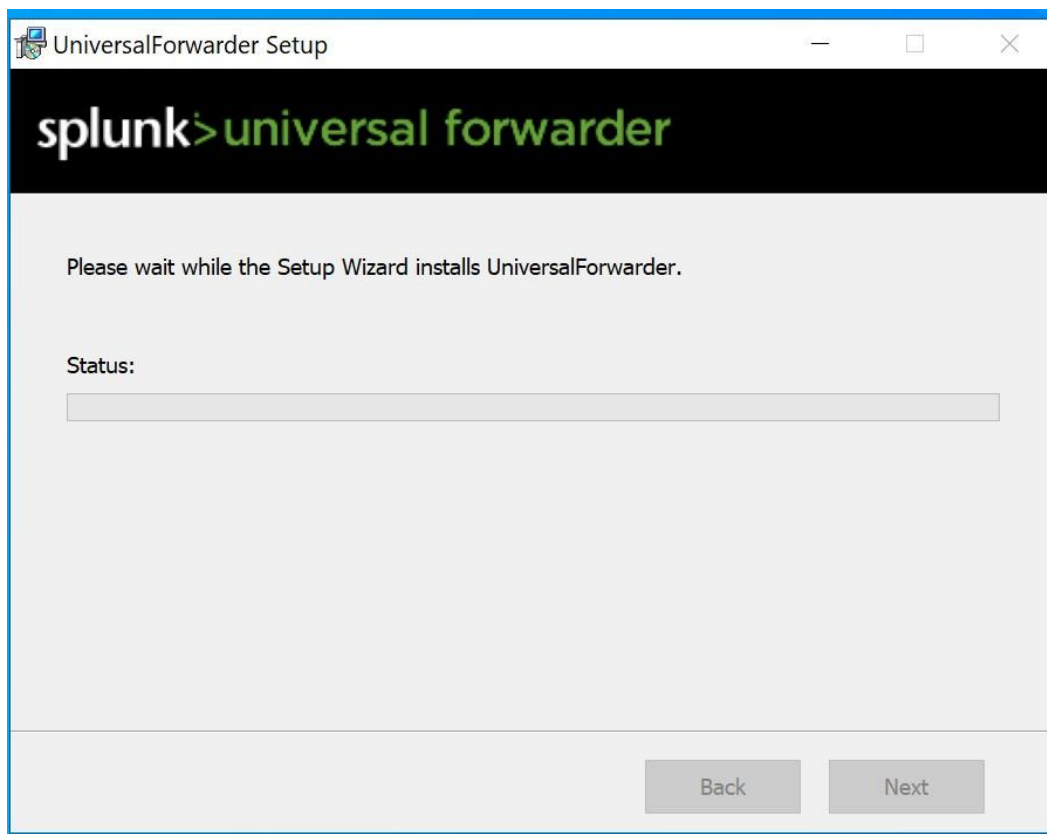


Figure 13. Universal Forwarder installation progress showing active component deployment and service registration on the endpoint.

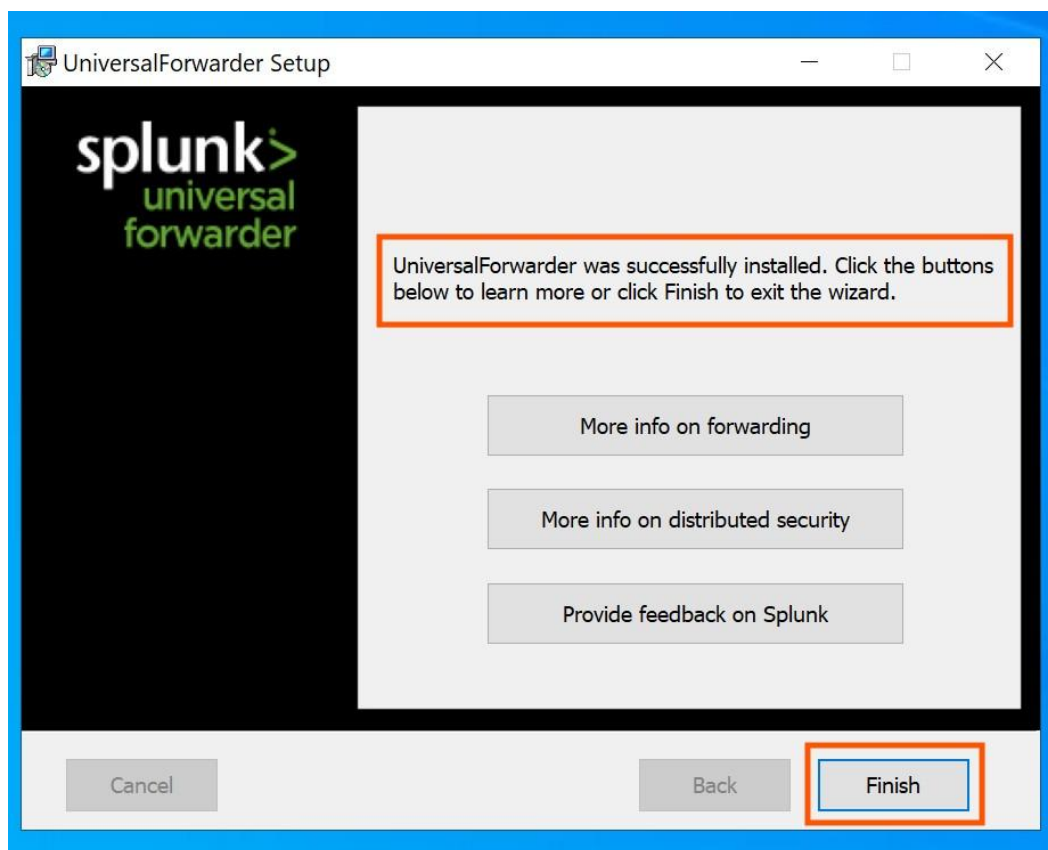
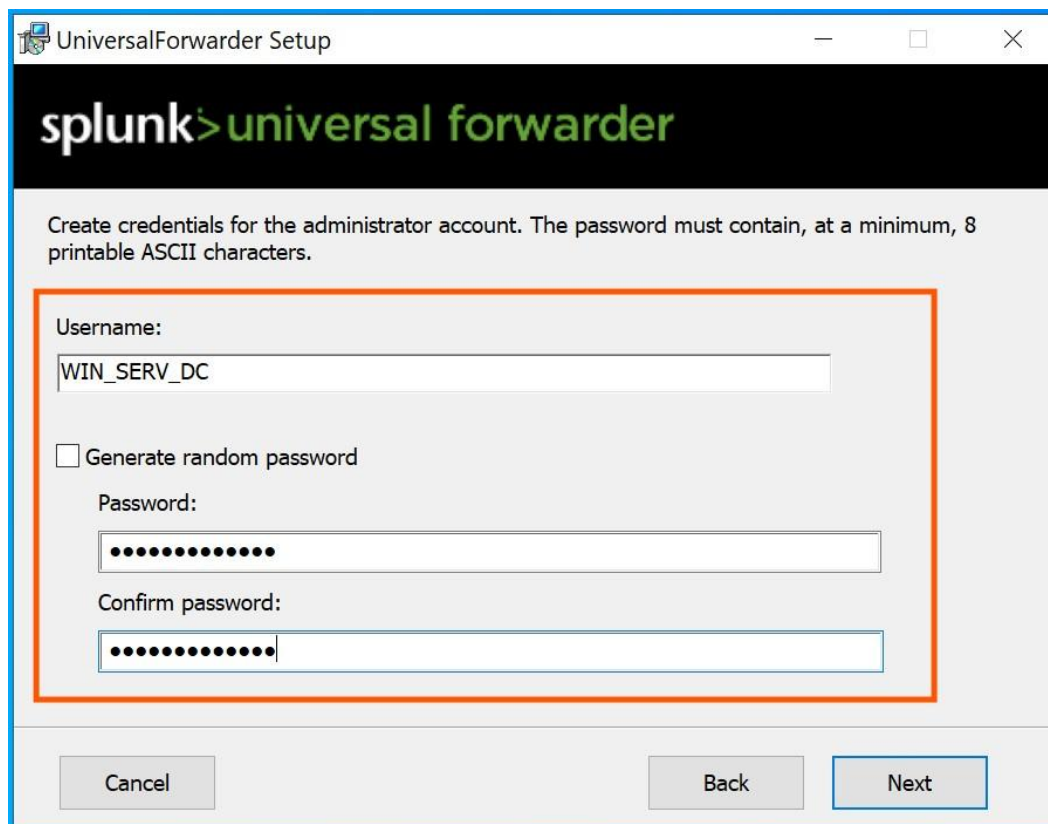


Figure 14. Successful completion of the Splunk Universal Forwarder installation. The setup wizard confirms that the UniversalForwarder was successfully deployed and is ready for configuration.

5.2.2 Forwarder Configuration

After installation, the Universal Forwarder was configured to connect to the Splunk Enterprise indexer. The configuration included specifying the receiving host IP address and the data collection inputs. The forwarder was configured to collect Windows Security, System, and Application event logs.

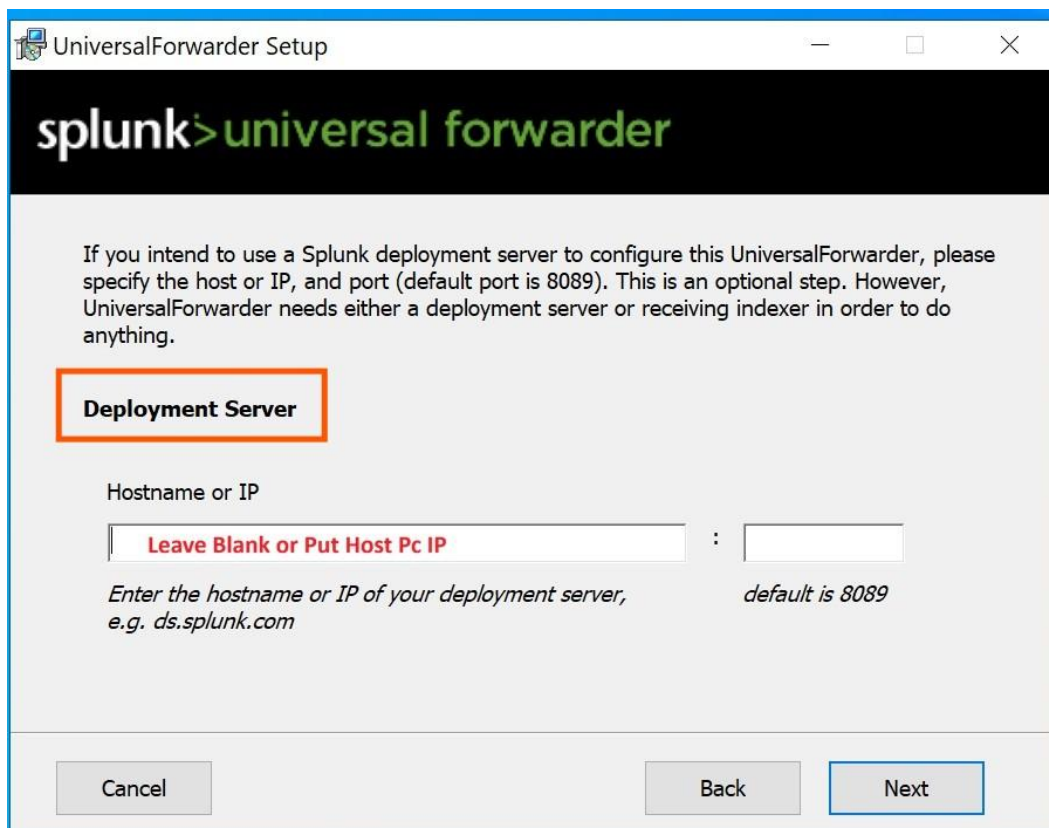


The screenshot shows the 'UniversalForwarder Setup' window. At the top, there's a header with the Splunk logo and 'universal forwarder' in green. Below the header, a message states: 'Create credentials for the administrator account. The password must contain, at a minimum, 8 printable ASCII characters.' The main form area is enclosed in an orange border and contains the following fields:

- Username:** A text box containing 'WIN_SERV_DC'.
- ☐ **Generate random password**
- Password:** A text box filled with 8 dots.
- Confirm password:** A text box filled with 8 dots.

At the bottom of the window, there are three buttons: 'Cancel', 'Back', and 'Next'.

Figure 15. Splunk Universal Forwarder configuration interface showing the receiving indexer connection parameters, including the target host IP address and receiving port (TCP 9997).

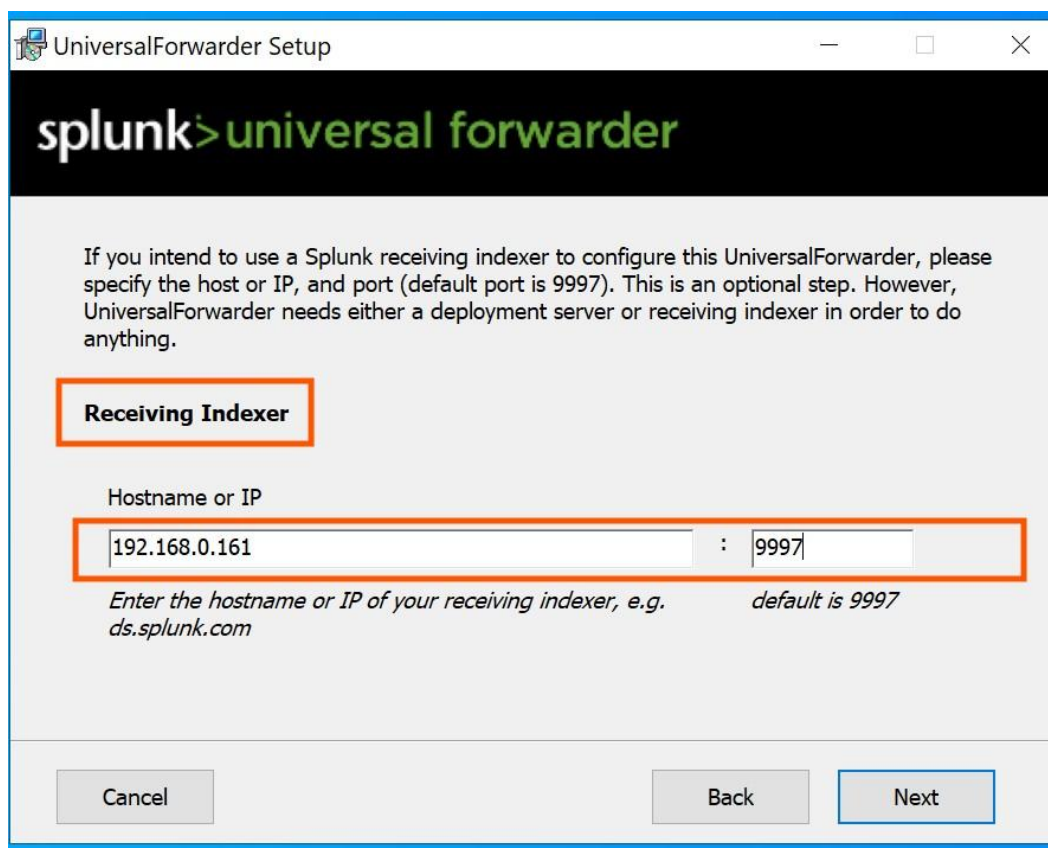


The image shows a Windows-style window titled "UniversalForwarder Setup". The window has a blue title bar with standard minimize, maximize, and close buttons. Below the title bar is a black header area with the "splunk" logo in white and "universal forwarder" in green. The main content area is light gray and contains the following text:

If you intend to use a Splunk deployment server to configure this UniversalForwarder, please specify the host or IP, and port (default port is 8089). This is an optional step. However, UniversalForwarder needs either a deployment server or receiving indexer in order to do anything.

Below this text is a section titled "Deployment Server" enclosed in an orange rectangular border. Under this section, the label "Hostname or IP" is followed by two input fields separated by a colon. The first input field contains the red text "Leave Blank or Put Host Pc IP". The second input field is empty. Below the input fields, there is a line of text: "Enter the hostname or IP of your deployment server, e.g. ds.splunk.com" on the left and "default is 8089" on the right. At the bottom of the window, there are three buttons: "Cancel", "Back", and "Next". The "Next" button is highlighted with a blue border.

Figure 16. Universal Forwarder host IP address configuration. The deployment host field was configured to establish the data forwarding path to the Splunk Enterprise indexer.



The screenshot shows the 'UniversalForwarder Setup' window. At the top, there's a header with the Splunk logo and the text 'splunk>universal forwarder'. Below this, a message states: 'If you intend to use a Splunk receiving indexer to configure this UniversalForwarder, please specify the host or IP, and port (default port is 9997). This is an optional step. However, UniversalForwarder needs either a deployment server or receiving indexer in order to do anything.'

The 'Receiving Indexer' section is highlighted with an orange border. It contains a label 'Hostname or IP' above a text input field. The input field is divided into two parts: the first part contains '192.168.0.161' and the second part contains '9997', separated by a colon. Below the input field, there is a hint: 'Enter the hostname or IP of your receiving indexer, e.g. ds.splunk.com' and 'default is 9997'.

At the bottom of the window, there are three buttons: 'Cancel', 'Back', and 'Next'. The 'Next' button is highlighted with a blue border.

Figure 17. Splunk indexer receiving configuration showing the host PC IP address mapping, establishing the data pipeline between the endpoint and the central SIEM server.

5.2.3 Forwarder Verification on Server

To confirm the Universal Forwarder was properly deployed on the monitored endpoint, the Windows Programs and Features control panel was accessed on the server to verify the installation. The hostname of the Windows Server was confirmed using the 'hostname' command in the Command Prompt, returning 'WIN-CJ0J0L41QG0'. This verification step ensured both the server and endpoint identities were correctly established in the deployment.

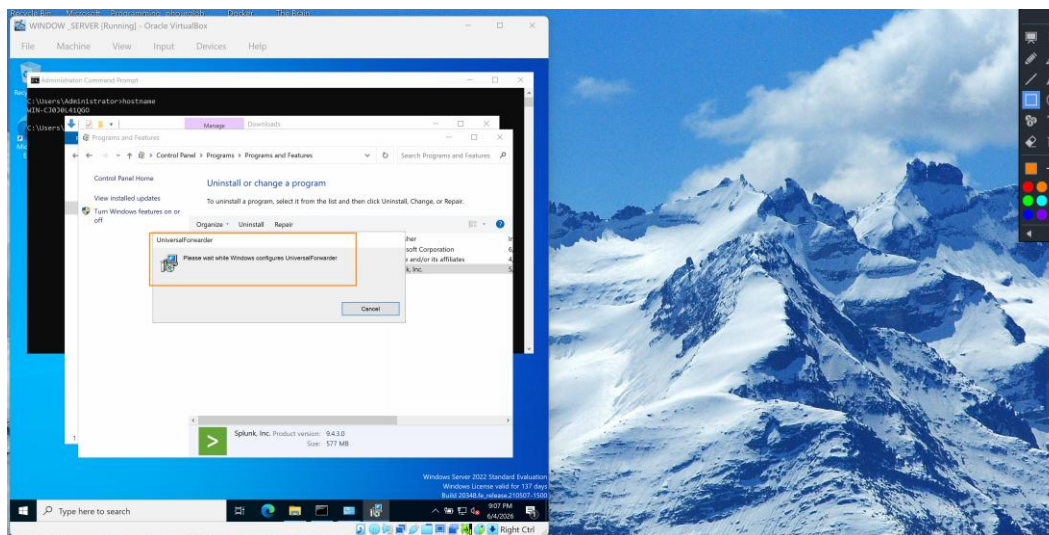


Figure 18. Windows Server Programs and Features panel showing the UniversalForwarder installation (version 9.4.3, 577 MB) being configured. The Command Prompt displays the server hostname 'WIN-CJ0J0L41QG0' confirming the server identity in the SIEM architecture.

5.3 Phase 3: Windows Firewall Configuration

A critical component of the SIEM deployment is the proper configuration of network firewall rules. Windows Defender Firewall with Advanced Security was configured on both the server and the monitored endpoint to permit bidirectional Splunk traffic. Without these rules, the Universal Forwarder would be unable to transmit log data to the indexer, and the Splunk web interface would be inaccessible from network clients.

5.3.1 Inbound Rule Configuration

Inbound firewall rules were created on the Splunk Enterprise server to allow incoming connections on the Splunk management port (TCP 8000) and the data receiving port (TCP 9997). The splunkd service executable was explicitly permitted through the firewall for all network profiles (Domain, Private, and Public).

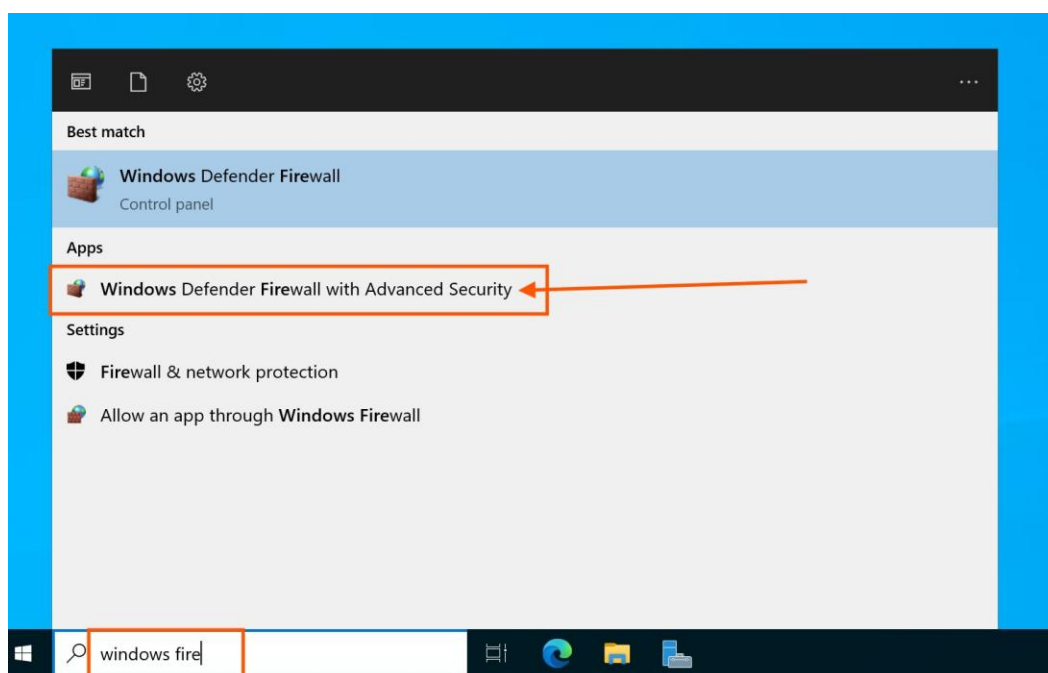


Figure 19. Windows Defender Firewall with Advanced Security management console opened on the server, displaying the interface for configuring inbound and outbound security rules.

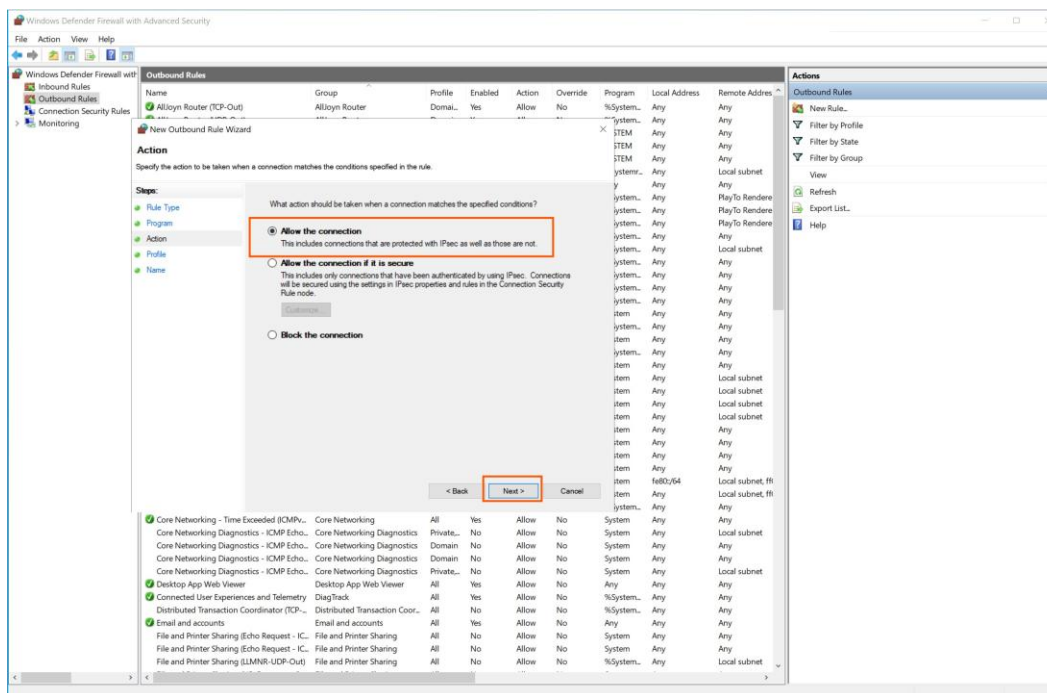


Figure 20. Firewall rule action configuration set to 'Allow the connection', permitting inbound Splunk traffic through the Windows Defender Firewall.

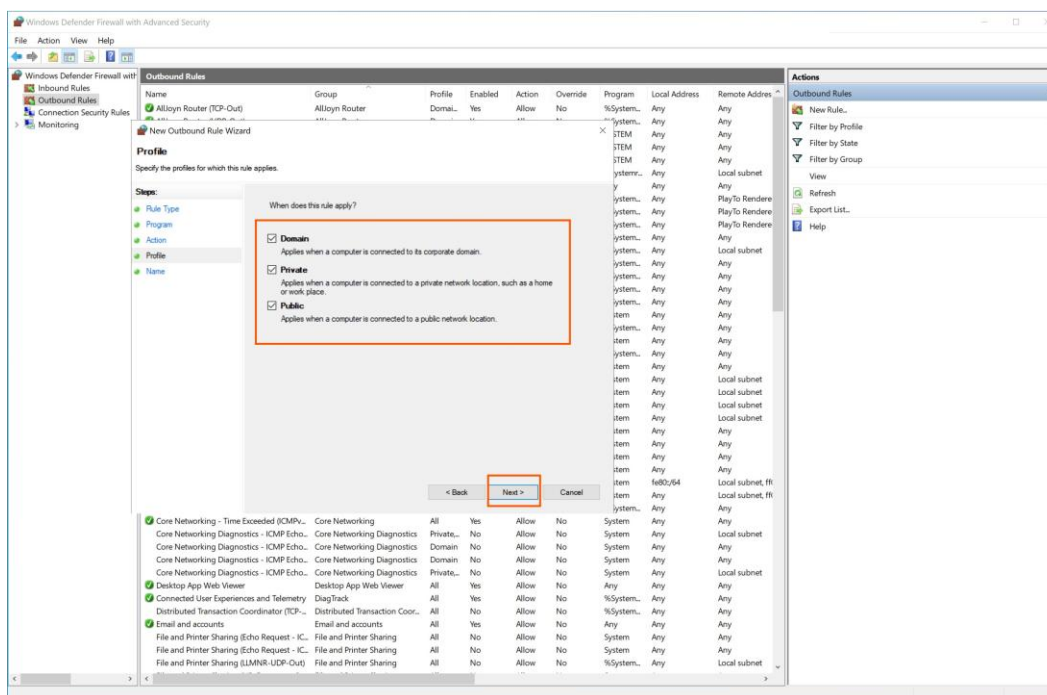


Figure 21. Network profile selection for the Splunk firewall rule, applying the 'Allow' action across all three profiles: Domain, Private, and Public networks, ensuring comprehensive accessibility.

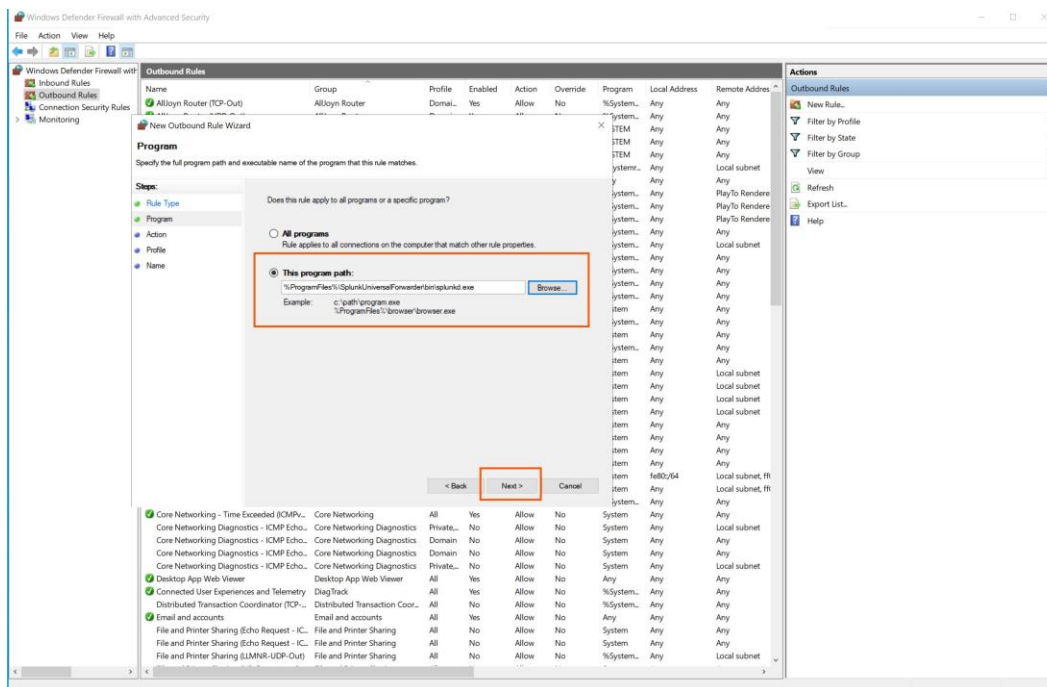


Figure 22. Confirmation of the splunkd firewall rule successfully added to the Windows Defender Firewall inbound rules list, permitting Splunk service communication.

5.3.2 Outbound Rule Configuration

Outbound firewall rules were configured on the monitored endpoint to allow the Universal Forwarder to transmit log data to the Splunk indexer. The New Outbound Rule Wizard was used to create a program-specific rule targeting the Splunk forwarder executable, permitting outbound connections to the indexer on the designated receiving port.

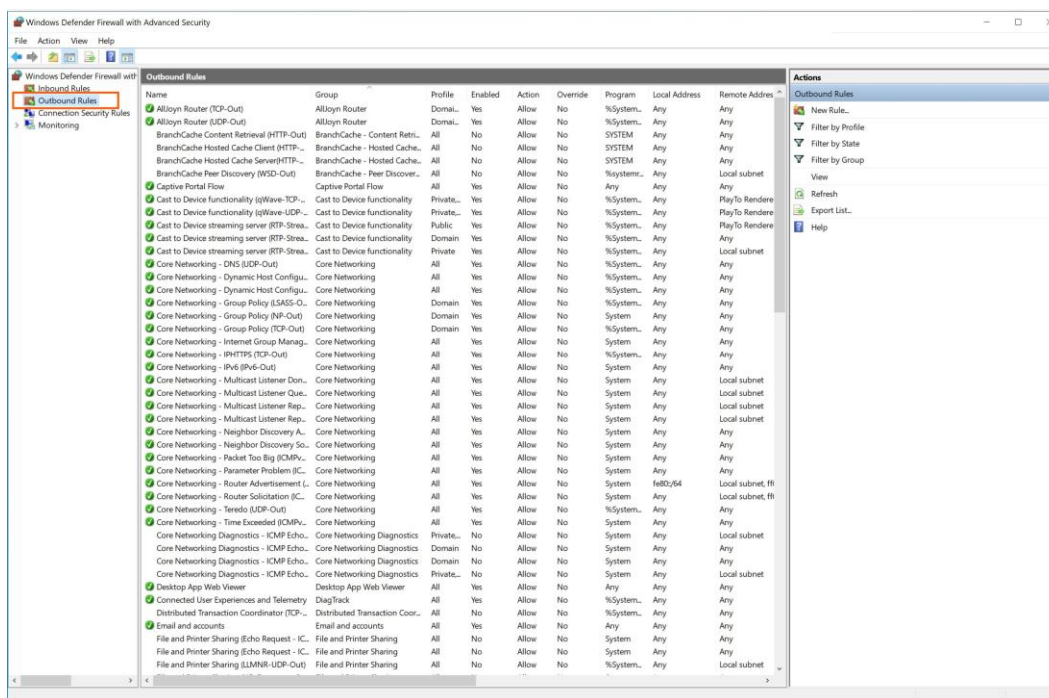


Figure 23. Windows Defender Firewall outbound rule configuration initiated to enable the Universal Forwarder to transmit logs through the configured receiving port to the Splunk indexer.

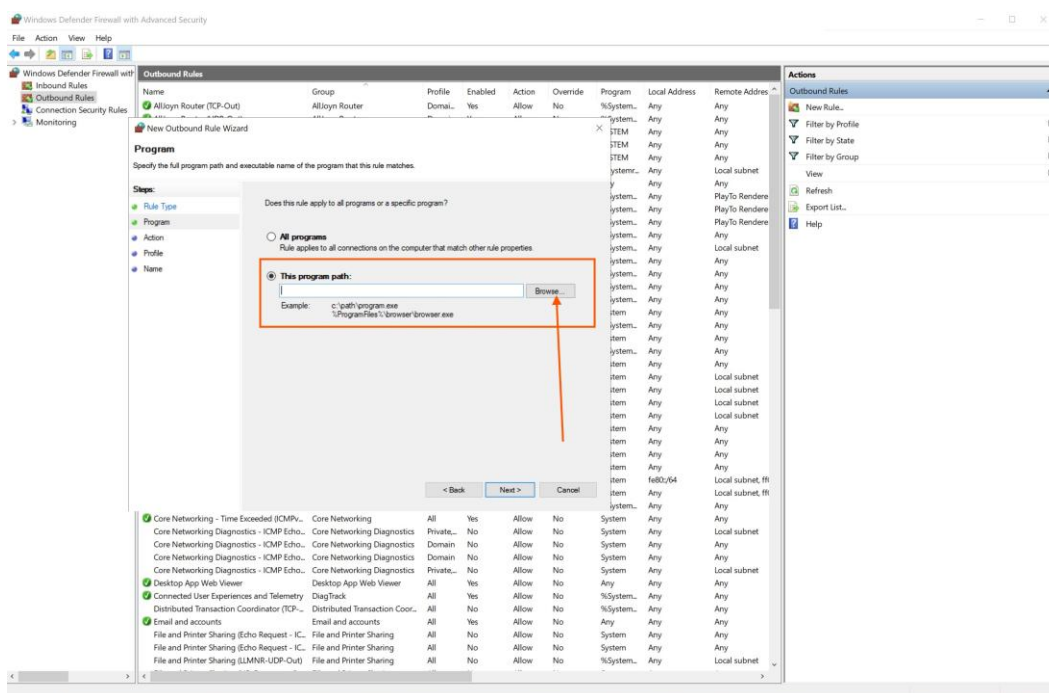


Figure 24. New Outbound Rule Wizard displaying the program path specification step. The rule targets the Splunk forwarder executable to permit outbound log transmission.

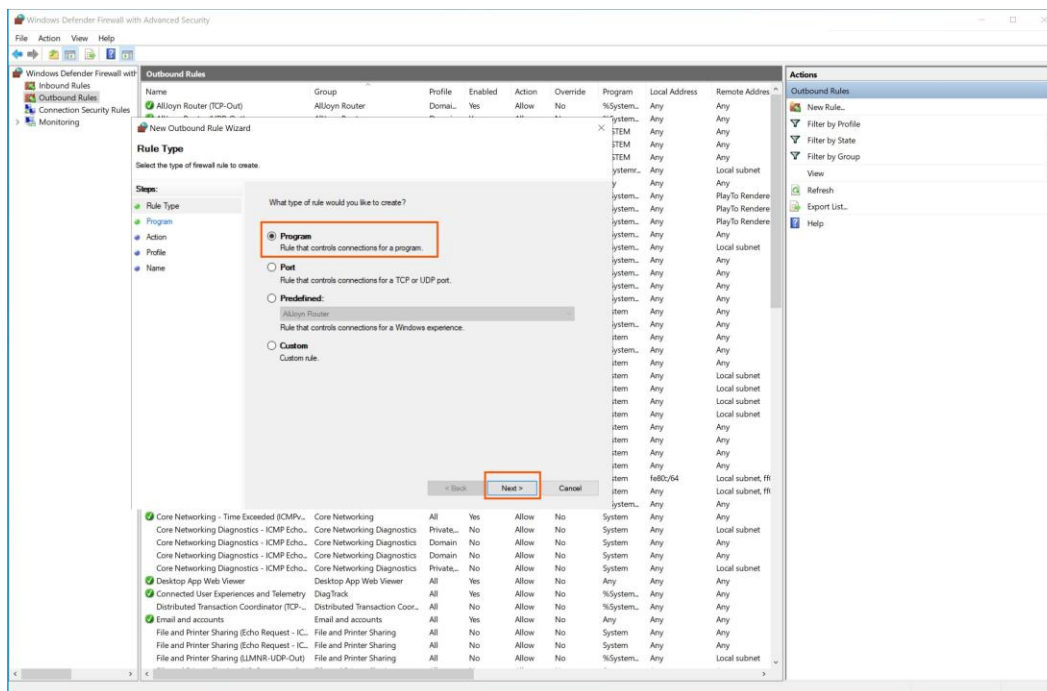


Figure 25. Outbound rule program selection interface in Windows Defender Firewall, allowing specification of the exact executable path for the Splunk Universal Forwarder.

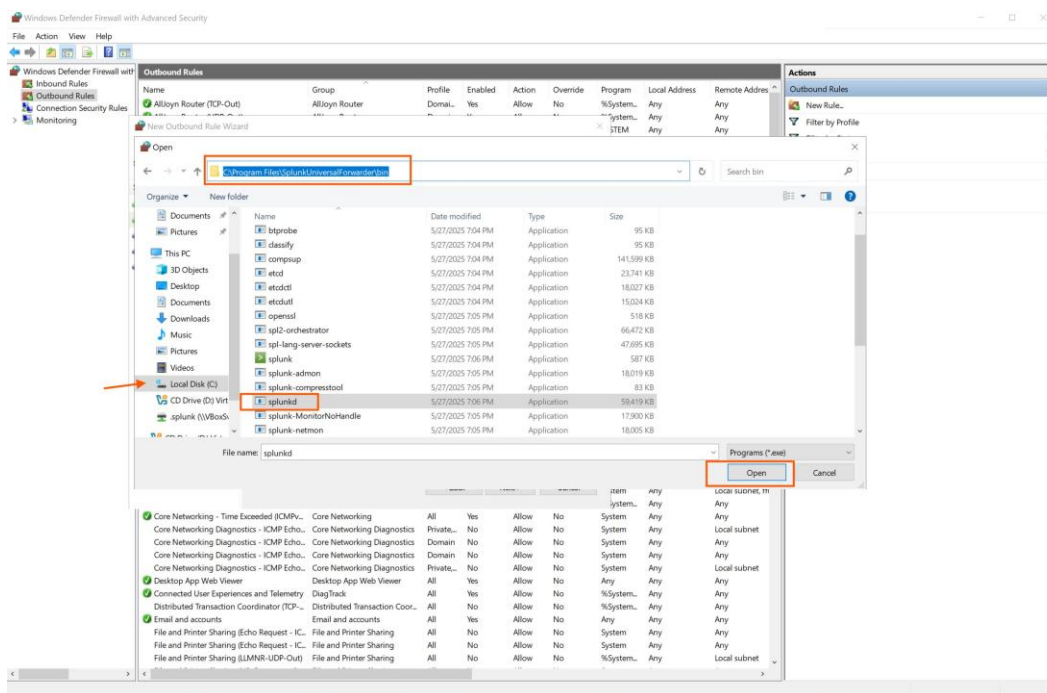


Figure 26. File system navigation to locate the Splunk Universal Forwarder executable for the firewall outbound rule program path configuration.

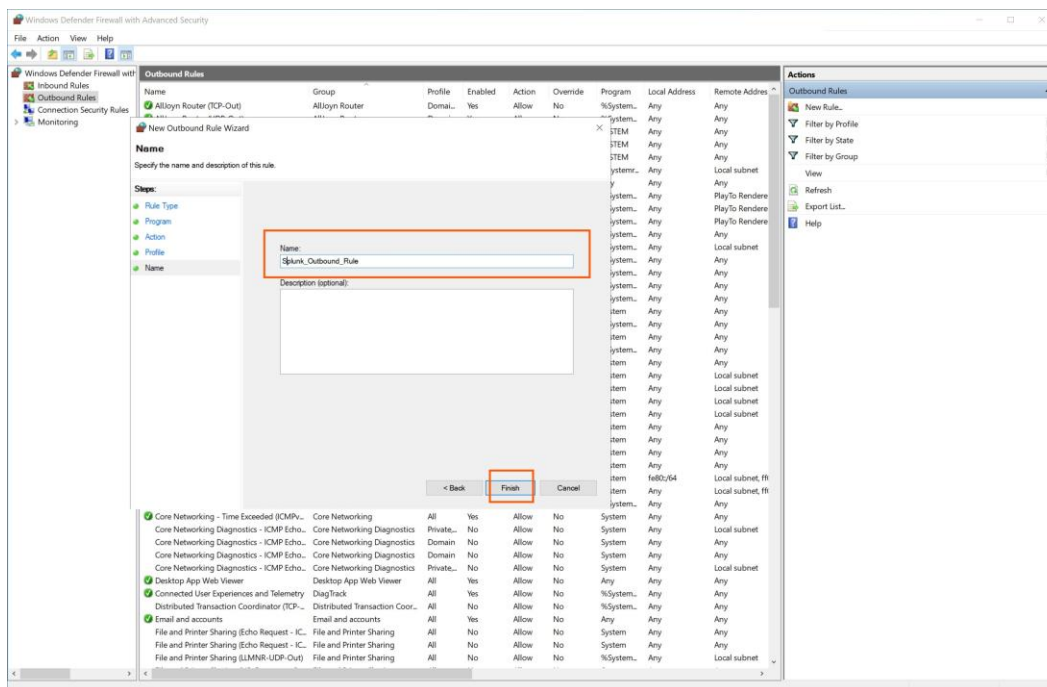


Figure 27. Naming the outbound firewall rule with a descriptive identifier for administrative clarity and future reference in the firewall rule management console.

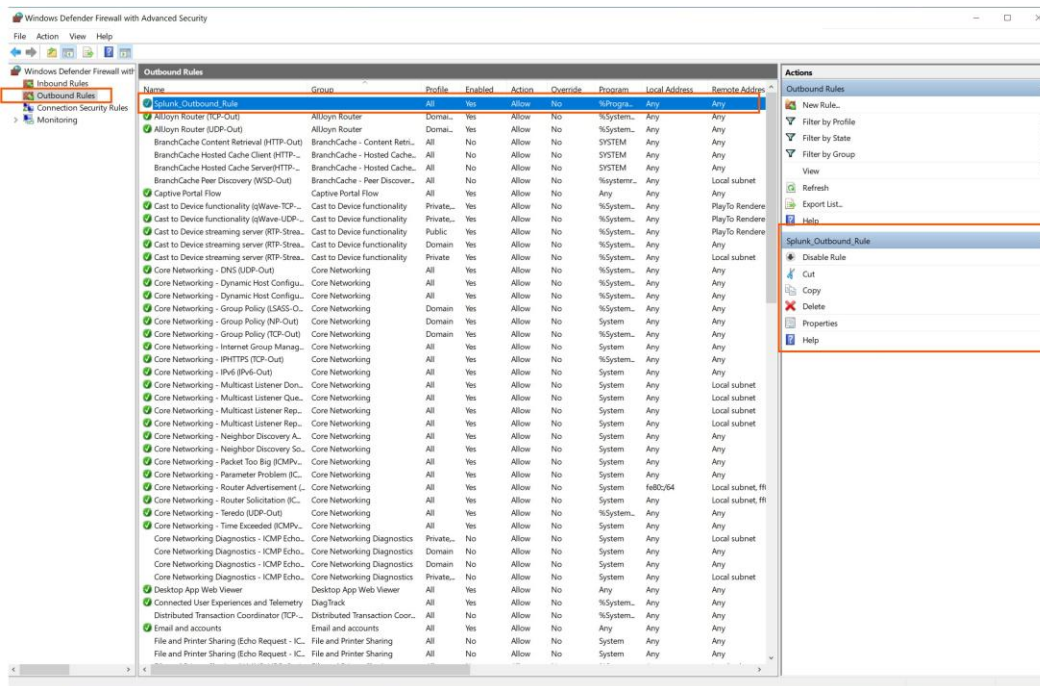


Figure 28. Confirmation that the Splunk outbound firewall rule has been successfully created and enabled in Windows Defender Firewall, permitting the Universal Forwarder to transmit log data to the indexer.

5.4 Phase 4: Indexer and Receiving Port Configuration

With the Universal Forwarder deployed and firewall rules in place, the Splunk Enterprise indexer was configured to listen on the designated receiving port (TCP 9997) for incoming data from the forwarder. This configuration establishes the data pipeline endpoint on the server side, enabling the indexer to accept, parse, and index forwarded log data.

The receiving configuration was completed through the Splunk Enterprise web interface under Settings > Forwarding and receiving > Configure receiving. A new receiving port was added (9997), and the indexer was configured to accept connections from all forwarders in the deployment.

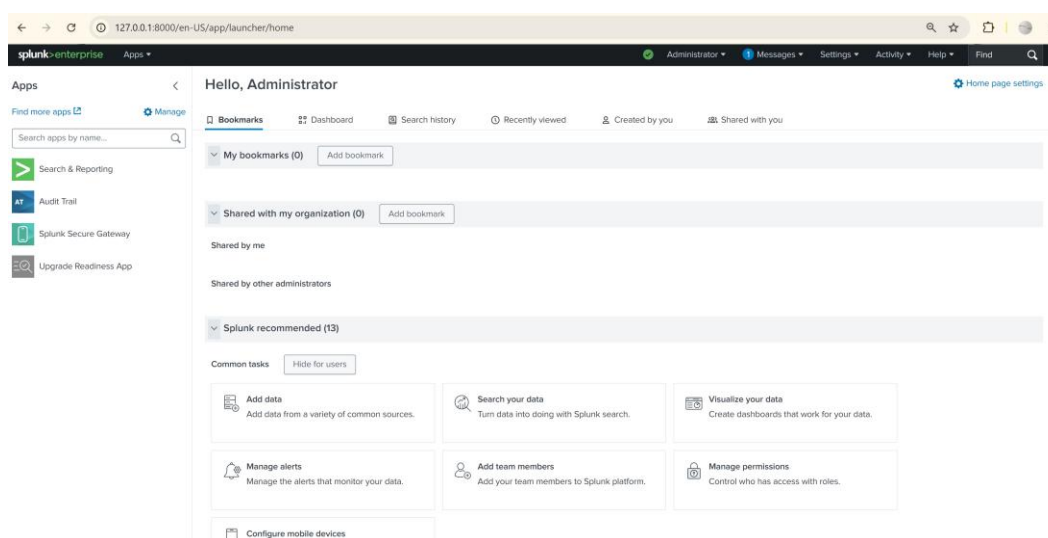


Figure 29. Splunk Enterprise Settings menu navigation to the Forwarding and Receiving configuration panel, where the data receiving port is established for incoming Universal Forwarder connections.

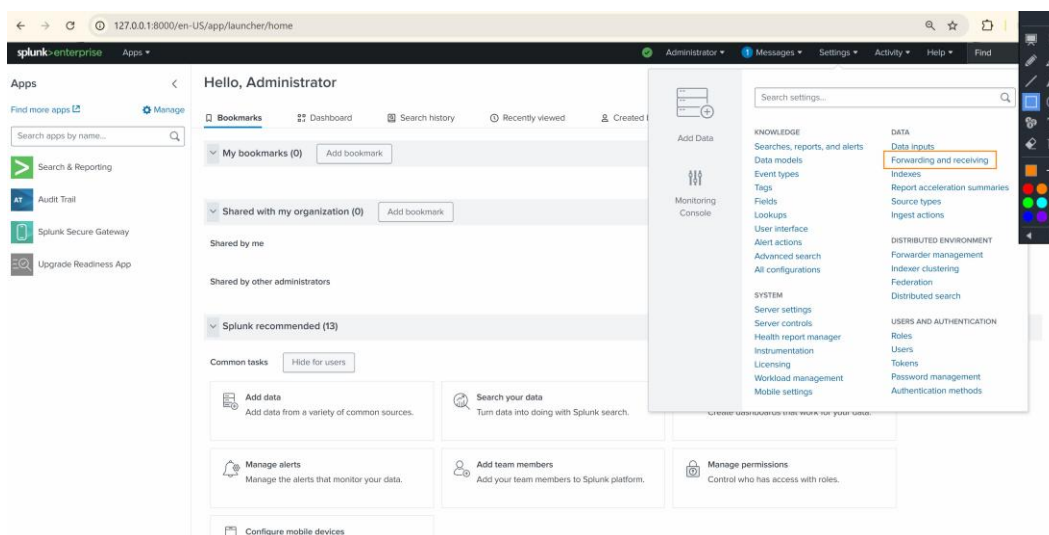


Figure 30. Splunk Enterprise receiving port configuration interface displaying the setup of TCP port 9997 as the designated data receiving endpoint for Universal Forwarder connections.

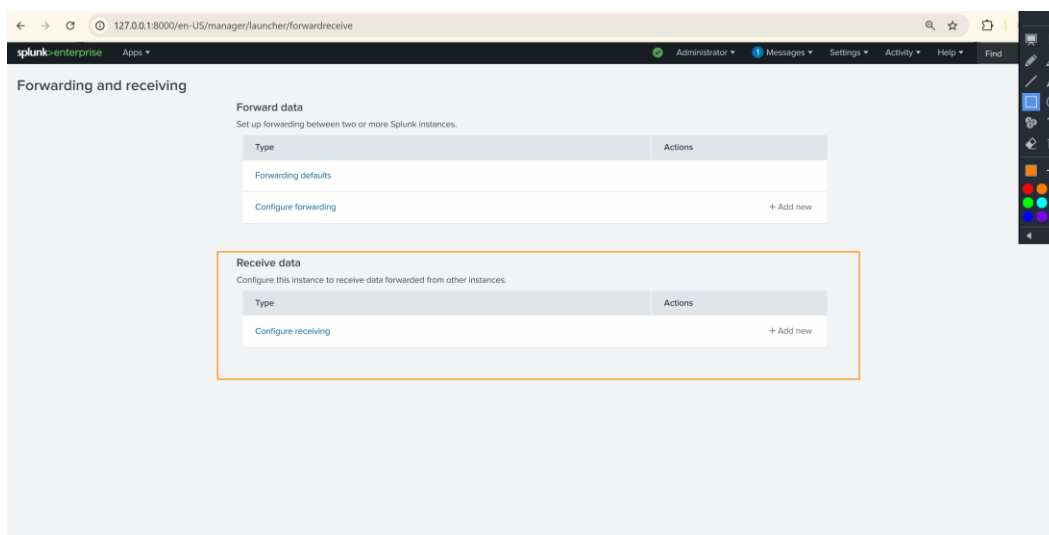


Figure 31. Confirmation of the receiving port (TCP 9997) successfully configured on the Splunk Enterprise indexer, ready to accept incoming forwarder data.

5.5 Phase 5: Data Ingestion Verification

Following the completion of all configuration steps, the data ingestion pipeline was verified by executing search queries on the Splunk Enterprise interface. The initial search query 'index=*' was used to retrieve all indexed events, confirming that the Universal Forwarder was successfully transmitting data to the indexer.

The verification confirmed that 5,406 events had been successfully ingested from the monitored endpoint (DESKTOP-EPLK670) covering a time range from June 3, 2026 to June 4, 2026. The events included Windows Security logs, System logs, and Setup logs, with multiple event sources including WinEventLog:Security and WinEventLog:Setup.

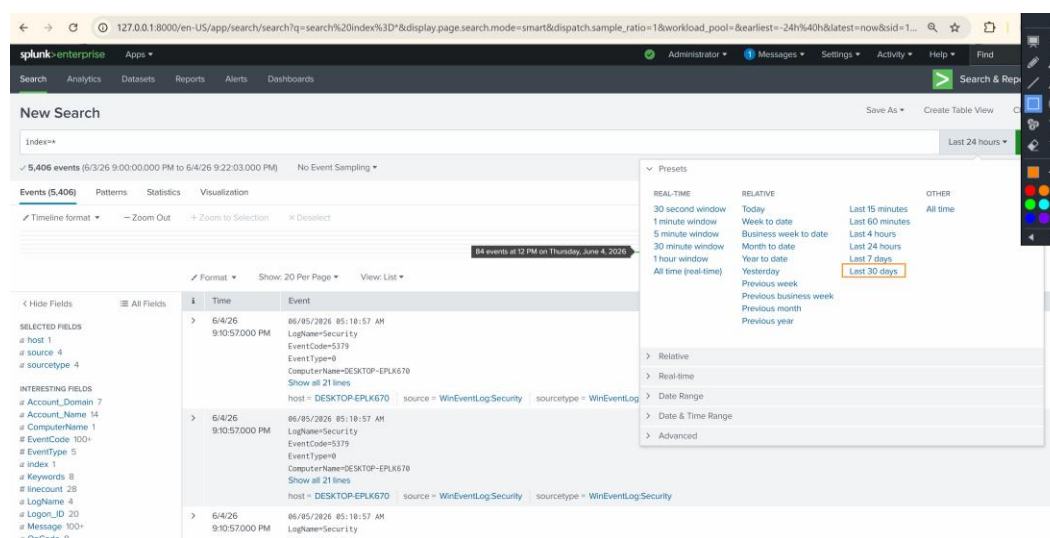


Figure 32. Splunk Enterprise search results confirming successful data ingestion. The 'index=*' query returned 5,406 events from the monitored endpoint DESKTOP-EPLK670. The time range picker overlay shows the available presets including Real-Time, Relative, and Date Range options for granular event filtering.

The search results revealed important metadata about the ingested events. The Selected Fields panel on the left showed 1 unique host (DESKTOP-EPLK670), 4 source types (WinEventLog:Security, WinEventLog:Setup, WinEventLog:System, and WinEventLog:Application), confirming comprehensive log collection across multiple Windows event channels.

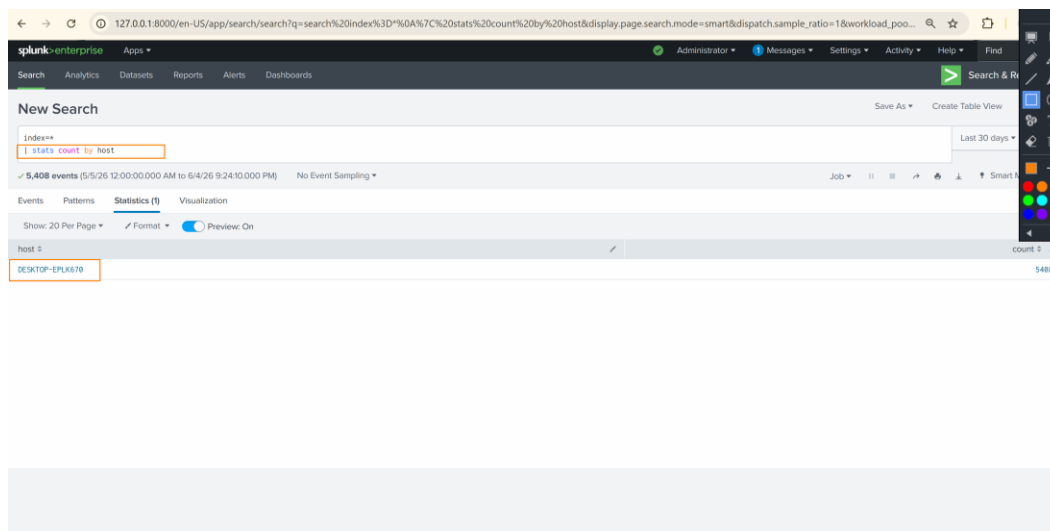


Figure 33. Detailed event inspection view showing individual security events with full metadata including LogName, EventCode, EventType, ComputerName, and source classification fields.

5.6 Phase 6: SPL Search and Analysis

With data ingestion confirmed, targeted SPL (Search Processing Language) queries were developed to extract actionable security intelligence from the indexed events. SPL is Splunk's proprietary query language that enables analysts to search, filter, transform, and visualize machine data for security monitoring and incident investigation purposes.

5.6.1 Host-Specific Event Filtering

The first analytical query targeted the specific monitored endpoint by filtering events using the host field. The query 'index=* host="DESKTOP-EPLK670" EventCode=1' was executed to isolate system setup events (EventCode 1) from the monitored Windows endpoint. This query returned 8 events within the Last 30 days time range, confirming active system configuration activities on the endpoint.

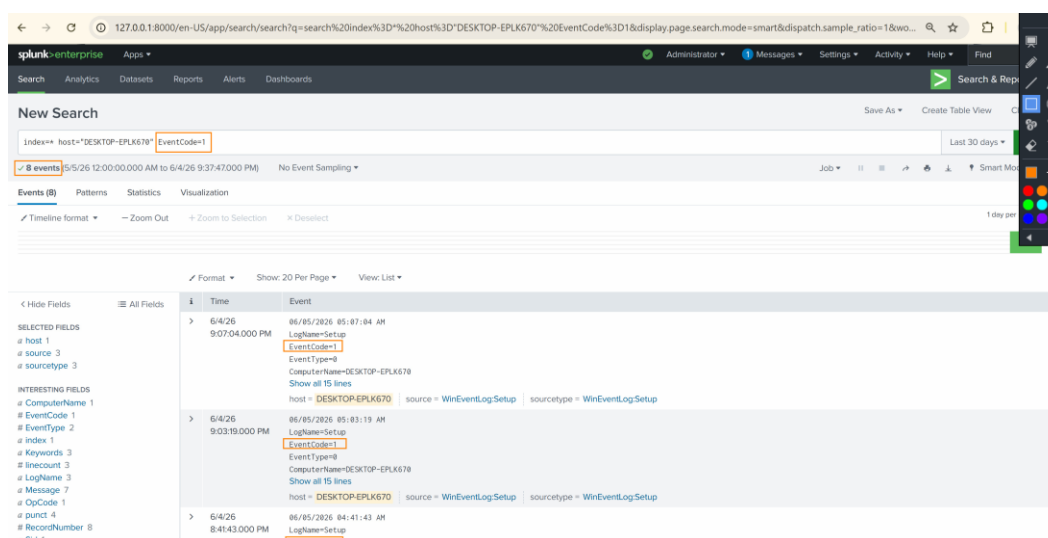


Figure 34. SPL query results for host-specific event filtering using 'index=* host="DESKTOP-EPLK670" EventCode=1'. The search returned 8 setup events (EventCode=1) from the WinEventLog:Setup source, demonstrating precise event filtering capabilities for targeted security analysis.

5.6.2 Security Event Analysis

Additional SPL queries were executed to analyze security-specific events, including failed logon attempts (EventCode 4625) and successful authentications (EventCode 4624). These queries provided visibility into authentication patterns, identifying potential unauthorized access attempts and establishing baselines for normal authentication behavior. The Interesting Fields

panel revealed valuable metadata distributions including Account_Domain (7 unique values), Account_Name (14 unique values), and EventCode (100+ unique values), providing a comprehensive view of the security event landscape.

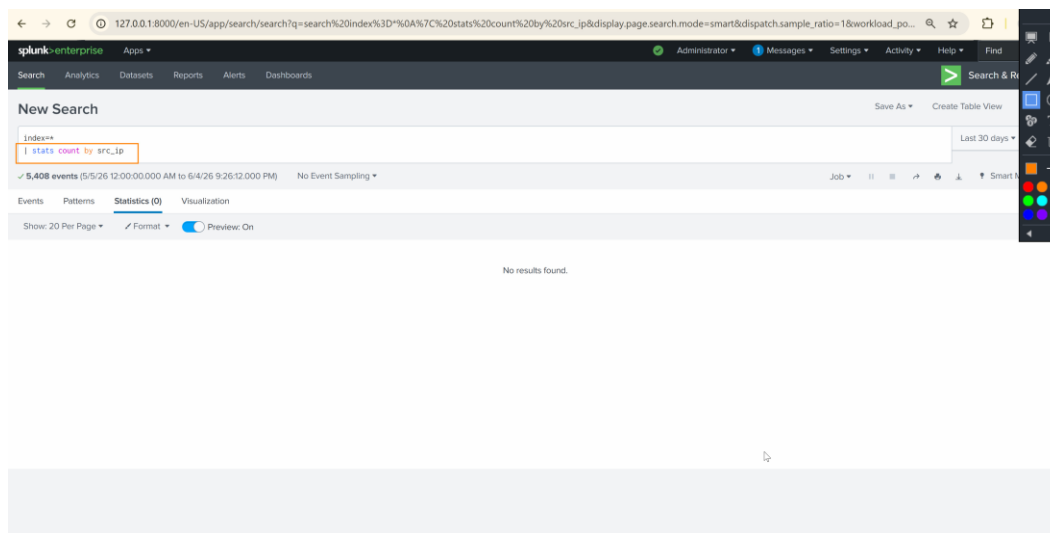


Figure 35. SPL search query execution targeting security event analysis with field extraction results showing the distribution of event codes, account names, and logon types across the indexed dataset.

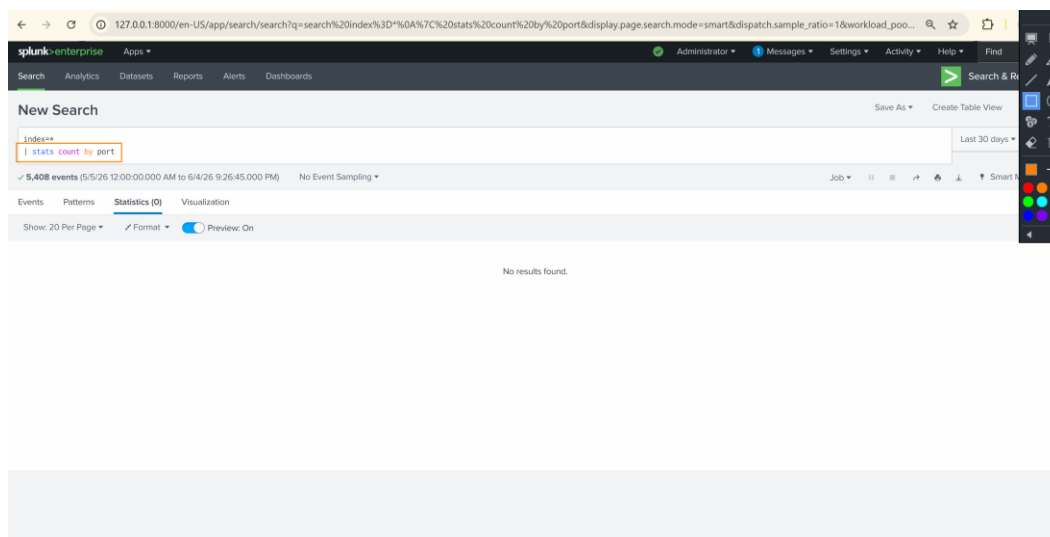


Figure 36. Expanded event detail view showing individual security event fields including Account_Name, LogonType, and associated metadata for authentication event correlation.

5.7 Phase 7: Security Dashboard Creation

The final phase of the implementation involved the creation of a custom security dashboard within Splunk Enterprise. The Authentication Dashboard was designed to provide SOC analysts with a centralized, visual interface for monitoring authentication events, tracking failed logon trends, and identifying potential security threats in real time.

5.7.1 Dashboard Design and Panel Configuration

The Authentication Dashboard was created using Splunk's Dashboard Editor (Classic Dashboards) and configured with multiple visualization panels. Each panel was powered by a specific SPL query designed to extract and visualize a distinct aspect of the authentication security landscape.

The dashboard included the following panels:

- **Failed Logon Trend (Line Chart):** A time-series visualization using the query 'index=main sourcetype="WinEventLog:Security" EventCode=4625 | timechart span=1d count by host' to track daily failed authentication attempts per host.
- **Total Failed Logons (Single Value):** A prominent numerical display showing the total count of failed logon events (EventCode 4625) across the monitoring period, providing immediate visibility into authentication security posture.
- **Successful Authentication Events:** A panel tracking EventCode 4624 events to establish baseline authentication patterns and identify anomalous logon activity.

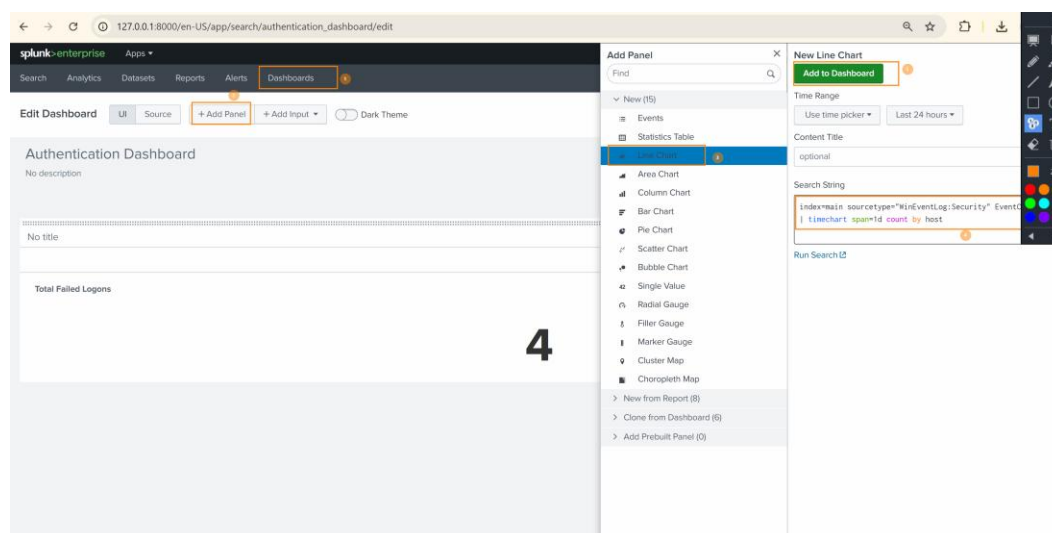


Figure 37. Splunk Dashboard Editor showing the Authentication Dashboard in edit mode. The Add Panel interface displays the Line Chart visualization type selected, with the SPL query 'index=main sourcetype="WinEventLog:Security" EventCode=4625 | timechart span=1d count by host' configured for the Failed Logon Trend panel. The Total Failed Logons single-value panel shows a count of 4 failed logon events.

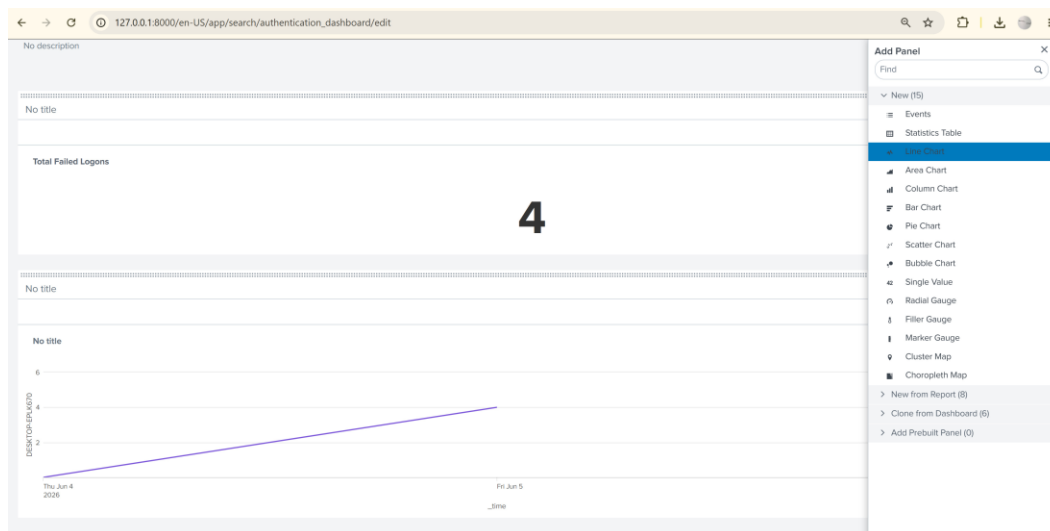


Figure 38. Authentication Dashboard panel configuration view showing the completed dashboard layout with the Failed Logon Trend line chart and Total Failed Logons counter panels positioned for optimal SOC analyst visibility.

5.7.2 Dashboard Panel SPL Queries

The following SPL queries were used to power the dashboard panels. Each query was optimized for performance and configured with appropriate time ranges to provide meaningful security insights.

Panel Name	SPL Query	Visualization
Failed Logon Trend	index=main sourcetype="WinEventLog:Security" EventCode=4625 timechart span=1d count by host	Line Chart
Total Failed Logons	index=main sourcetype="WinEventLog:Security" EventCode=4625 stats count	Single Value

Setup Events by Host	index=* host="DESKTOP-EPLK670" EventCode=1 timechart count	Column Chart
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6. Results and Findings

The Splunk SIEM deployment achieved all stated objectives, demonstrating a fully functional security monitoring pipeline from endpoint log collection through centralized indexing and visual dashboard reporting. The following key results were documented during the implementation.

6.1 Data Ingestion Metrics

Metric	Value
Total Events Ingested	5,406
Monitored Hosts	1 (DESKTOP-EPLK670)
Log Sources	4 (Security, System, Setup, Application)
Time Range	June 3, 2026 - June 5, 2026
Failed Logon Events (4625)	4
Setup Events (EventCode 1)	8
Unique Account Names	14
Unique Account Domains	7

6.2 Key Security Findings

The analysis of ingested security events revealed several noteworthy findings relevant to the security posture of the monitored environment:

- **Failed Authentication Attempts:** Four failed logon events (EventCode 4625) were identified during the monitoring period. These events were analyzed to determine the source accounts, logon types, and failure reasons, providing insight into potential unauthorized access attempts.
- **Event Distribution:** The EventCode field contained over 100 unique values, indicating a diverse range of system and security activities on the monitored endpoint. This breadth of event coverage demonstrates the comprehensive nature of the Windows event log collection.

- **Source Type Coverage:** The Universal Forwarder successfully collected logs from multiple Windows event channels, including Security, System, Setup, and Application logs, ensuring holistic visibility into endpoint activities.
- **Host Identification:** All events were correctly attributed to the source host DESKTOP-EPLK670, confirming accurate host identification in the forwarding pipeline.

7. Security Considerations

The following security considerations were addressed during the implementation to ensure the integrity and confidentiality of the SIEM deployment:

- **Firewall Hardening:** Windows Defender Firewall rules were configured with the principle of least privilege, permitting only the specific Splunk executables (splunkd.exe) through the firewall rather than opening broad port ranges.
- **Authentication Security:** A dedicated administrator account (phour44) was created for Splunk management, avoiding the use of default credentials. The account was configured with a strong password during installation.
- **Network Segmentation:** The lab environment utilized VirtualBox internal networking to simulate network segmentation between the SIEM server and monitored endpoints.
- **Data Integrity:** The Splunk Universal Forwarder establishes a persistent, reliable TCP connection to the indexer, ensuring that log data is transmitted without loss or tampering.
- **Encryption Considerations:** In a production deployment, SSL/TLS encryption should be configured between the Universal Forwarder and the indexer to protect log data in transit. This lab deployment focused on functional validation without encryption to simplify initial configuration.

8. Challenges and Lessons Learned

Several challenges were encountered during the implementation that provided valuable learning opportunities and insights applicable to future SIEM deployments:

- **Firewall Rule Ordering:** Initial data forwarding attempts failed because the Windows Defender Firewall was blocking outbound connections from the Universal Forwarder. This was resolved by creating specific outbound rules for the splunkd.exe executable. This experience reinforced the importance of comprehensive firewall auditing during SIEM deployments.
- **Universal Forwarder Hostname Resolution:** The Universal Forwarder initially had difficulty resolving the Splunk indexer hostname. This was resolved by using the direct IP address in the forwarder configuration, bypassing DNS dependencies in the lab environment.
- **Event Volume Management:** The initial 'index=*' query returned all indexed events without filtering, demonstrating the importance of developing targeted SPL queries for efficient analysis. The transition from broad queries to specific EventCode and host-based filters significantly improved analysis efficiency.
- **Dashboard Panel Optimization:** Creating effective dashboard panels required iterating on SPL queries to balance data granularity with visualization clarity. The timechart command with 'span=1d' aggregation proved effective for trend analysis in the Authentication Dashboard.

9. Conclusion and Recommendations

This project successfully demonstrated the end-to-end deployment of Splunk Enterprise as a SIEM solution for enterprise network monitoring. The implementation validated the complete log collection pipeline, from endpoint event generation through Universal Forwarder transmission, firewall traversal, indexer ingestion, and security dashboard visualization.

The deployment achieved all stated objectives: Splunk Enterprise was installed and configured as the central SIEM platform, the Universal Forwarder was deployed on a monitored endpoint, Windows Defender Firewall rules were established for bidirectional Splunk communication, data ingestion was verified with over 5,400 events, targeted SPL queries were developed for security analysis, and a custom Authentication Dashboard was created for real-time monitoring.

9.1 Recommendations for Production Deployment

- **Enable SSL/TLS Encryption:** Configure certificate-based encryption between Universal Forwarders and the indexer to protect log data in transit against interception and tampering.
- **Implement Role-Based Access Control (RBAC):** Define granular user roles and permissions within Splunk to enforce the principle of least privilege for SOC analysts, administrators, and read-only users.
- **Deploy Additional Forwarders:** Expand the monitoring footprint by deploying Universal Forwarders across all network endpoints, including servers, workstations, and network devices.
- **Configure Alerting:** Implement Splunk alerting rules for critical security events, such as brute force detection (multiple failed logons from a single source), privilege escalation events, and unusual logon patterns.
- **Integrate Threat Intelligence:** Leverage Splunk Enterprise Security (ES) or third-party threat intelligence feeds to correlate security events with known indicators of compromise (IOCs).
- **Establish Log Retention Policies:** Define data retention policies aligned with regulatory requirements (e.g., PCI DSS, HIPAA, SOX) to manage storage costs while maintaining compliance.

- **Implement Syslog Forwarding:** Extend data collection to include syslog sources from network infrastructure devices (firewalls, routers, switches) for comprehensive network visibility.

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11. Appendix: Figure Index

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